

CSE490 (D) INTRODUCTION TO FOOTBALL DATA AND ANALYTICS

IN ASSOCIATION WITH BANGLADESH FOOTBALL FEDERATION

SPRING 2026



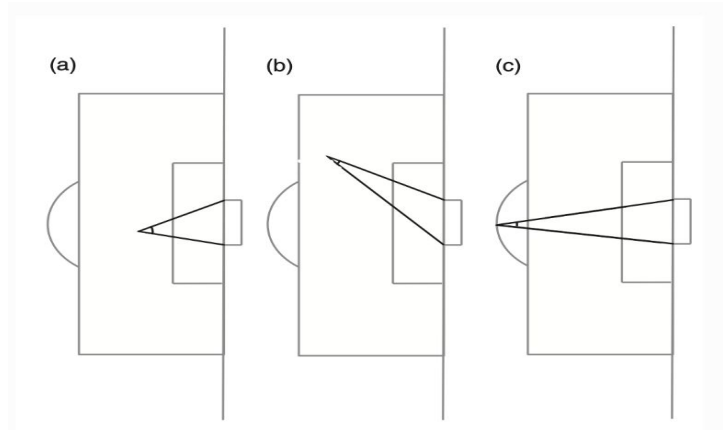
- Professional Certificate from BFF
- Internship Opportunity at **BFF Data Science** Department after successful completion
- Learning from the **industry experts** of Football Data Science
- Opportunity to **co-author** research papers for **Sports Data Conferences**

CSE490 (D) Introduction to Football Data & Analytics Lecture-4

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Expected Goals (xG)

- Which shot is most likely to be a goal?
- Measure the quality of a chance created.
- When we say that a shot had an xG of 0.2 (or 20%), we mean that typically shots of this type will be scored 20% of the time
- Which features do you think affect expected goals?



Expected Goals (xG)

- We want to predict what is going to happen
- Modeling captures our understanding of football
- Can you guess the outcome from this picture?



Expected Goals (xG) –factors

- **Shot Location**
 - Distance to goal
 - Angle to goal
- **Body Part Used**
 - Strong/weak foot
 - Header vs foot
- **Type of Assist**
 - Through ball, cross, cutback, rebound
- **Defensive & Goalkeeper Context**
 - Defensive pressure
 - Number/position of defenders
 - Goalkeeper positioning
- **Game Situation**
 - Open play vs set piece
 - Counterattack
 - Penalty / one-on-one



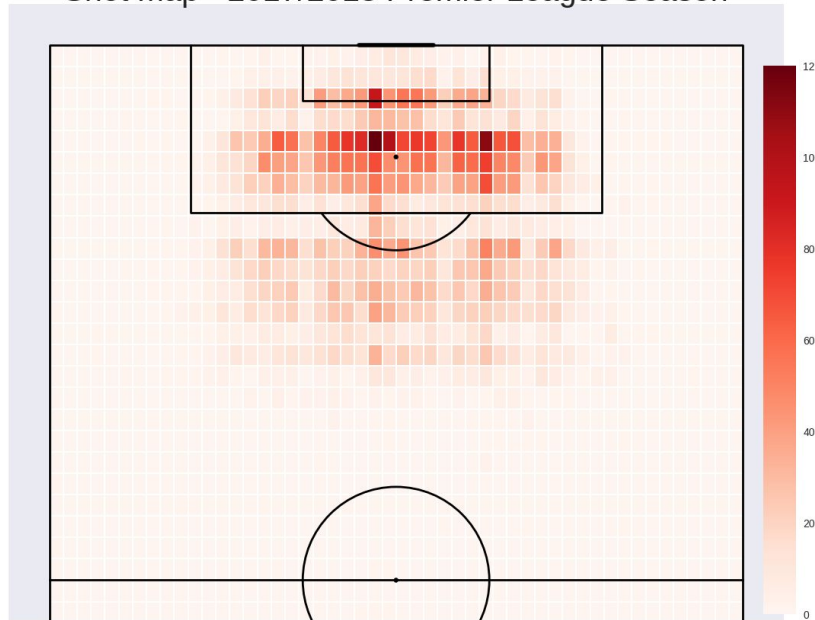
Expected Goals (xG)

- Do you think the team with the highest xG is the worthy winner ?
- xG tells us little more about the scoreline than the result itself.
- Take, for instance, a hard-fought match in the Champions League qualifiers between Bayern Munich and Lyon. Bayern won 1-0, but recorded 0.45-1.68 in xG.



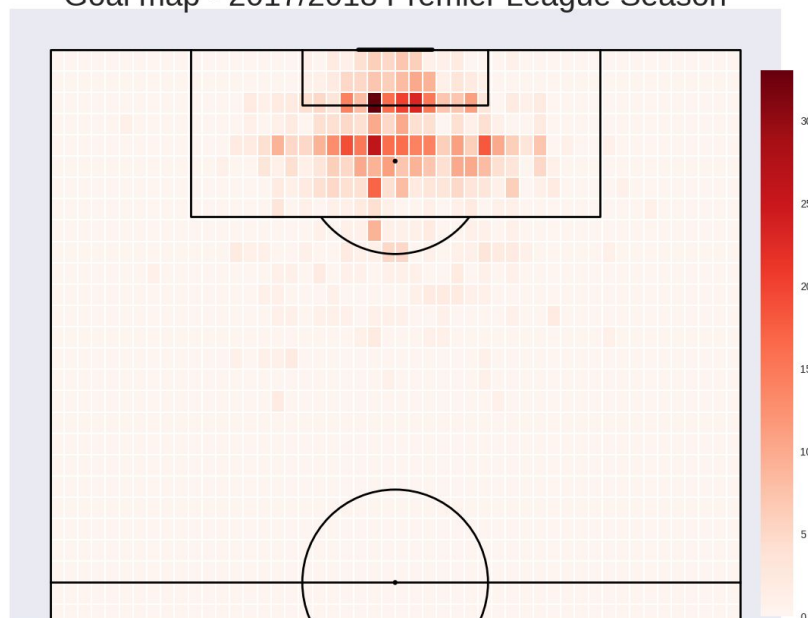
Shot Map

Shot map - 2017/2018 Premier League Season

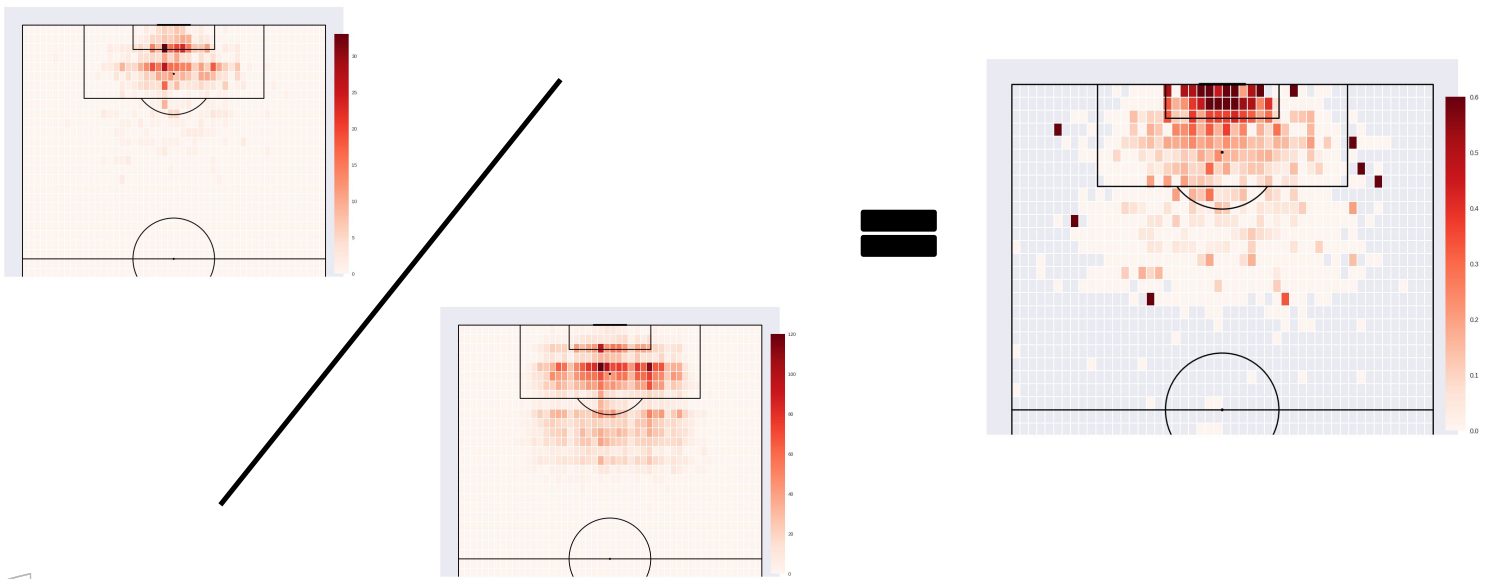


Goal Map

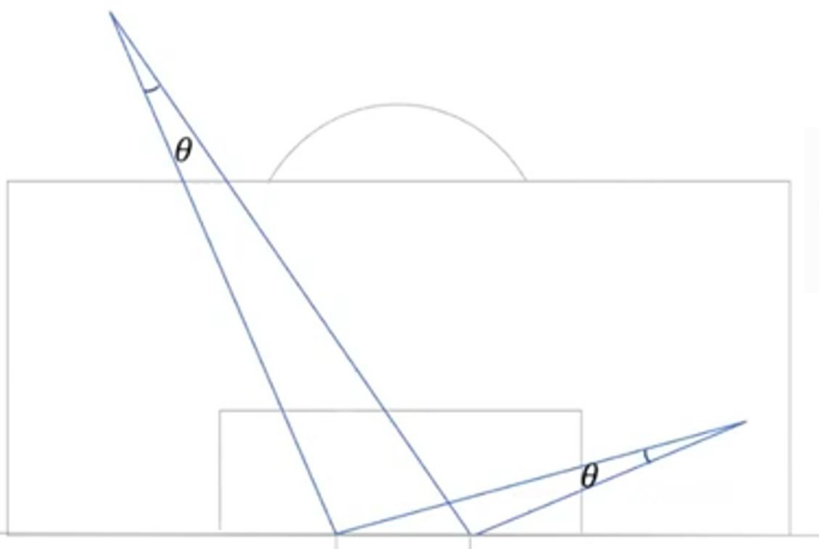
Goal map - 2017/2018 Premier League Season



Expected Goals (xG)

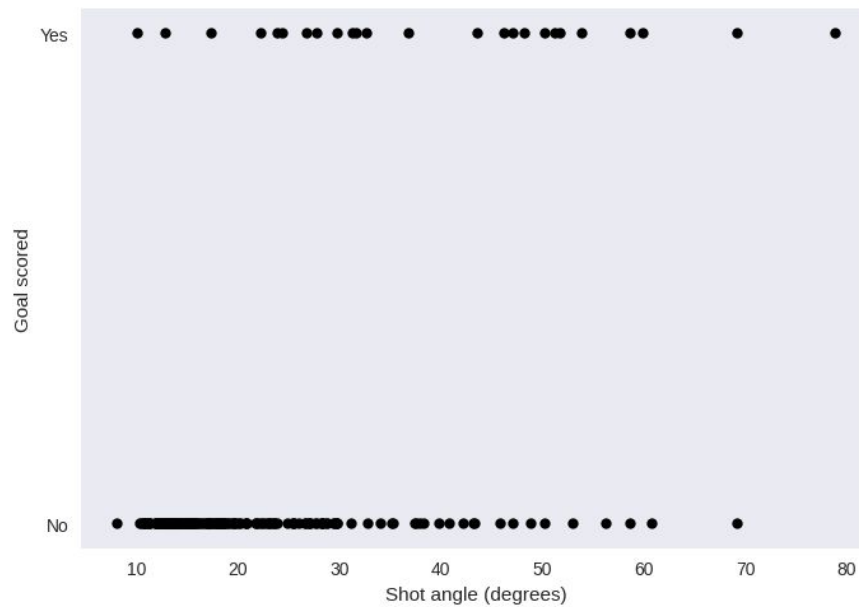


Modeling with angle



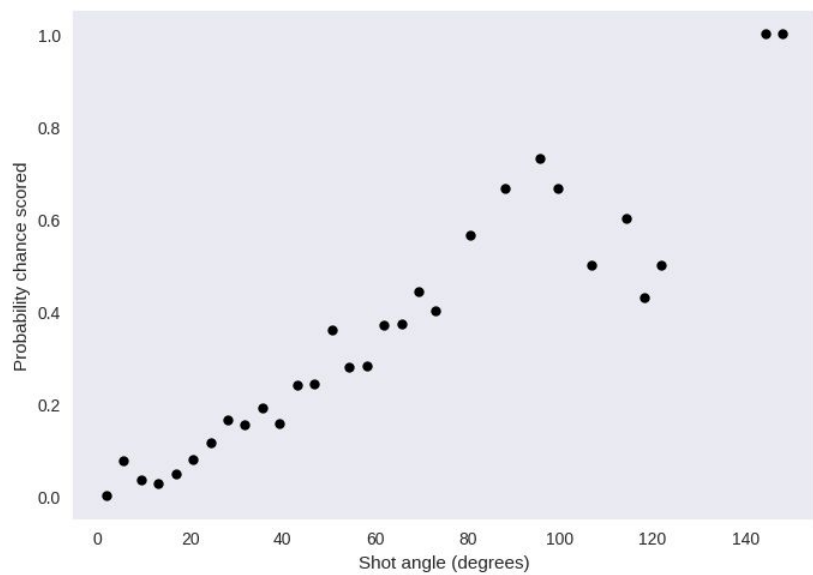
$$\tan(\theta) = \frac{7.32x}{x^2 + y^2 - (7.32/2)^2}$$

Modeling with angle



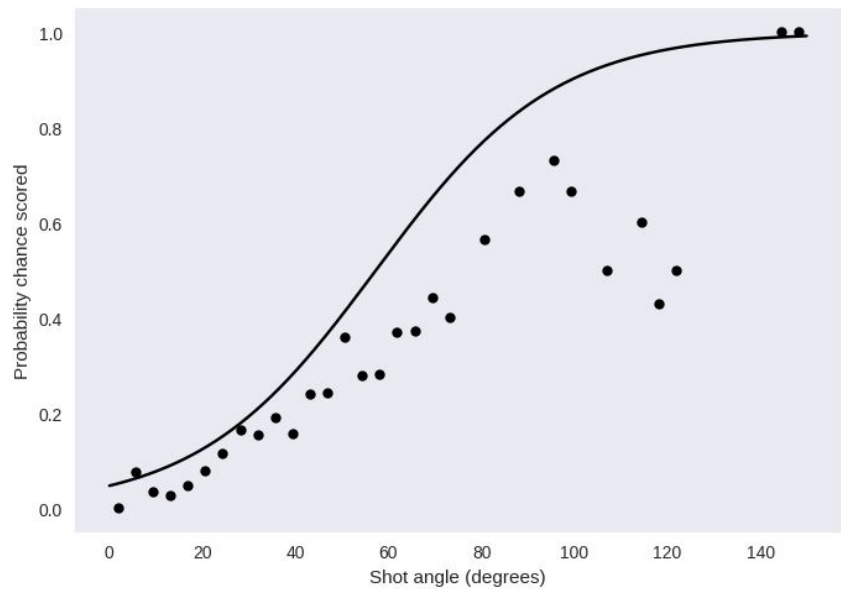
Modeling with angle

Can we fit a linear model here?



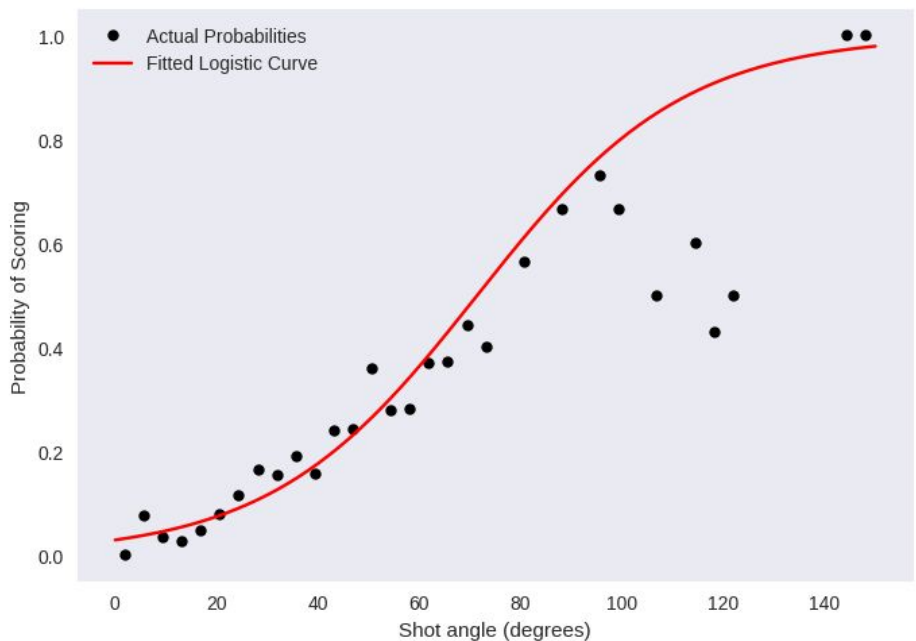
Modeling with angle

With randomly chosen
Parameters
 $b_0 = -3, b_1 = 3$

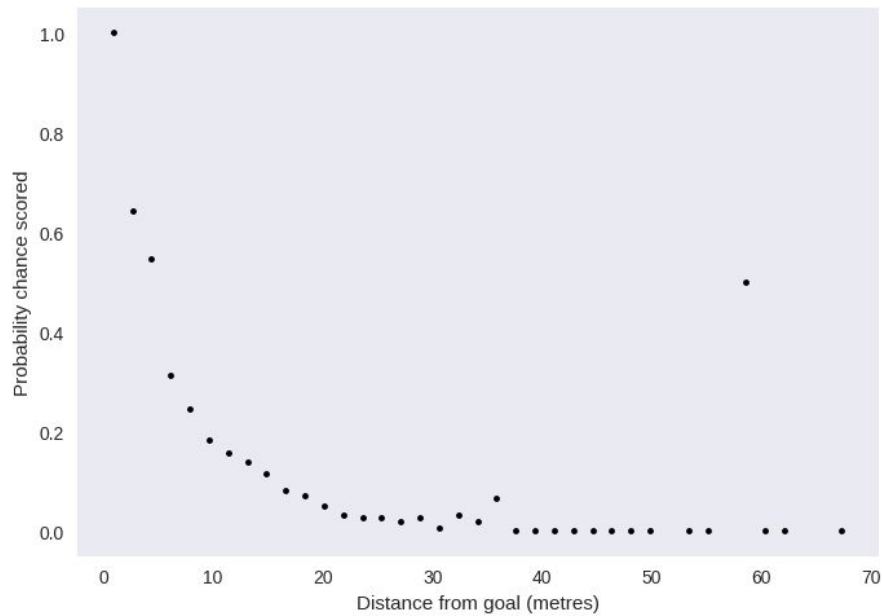


Modeling with angle

With logistic regression
Trained parameters
 $b_0 = -3.5069, b_1 = 2.8123$



Modeling with distance

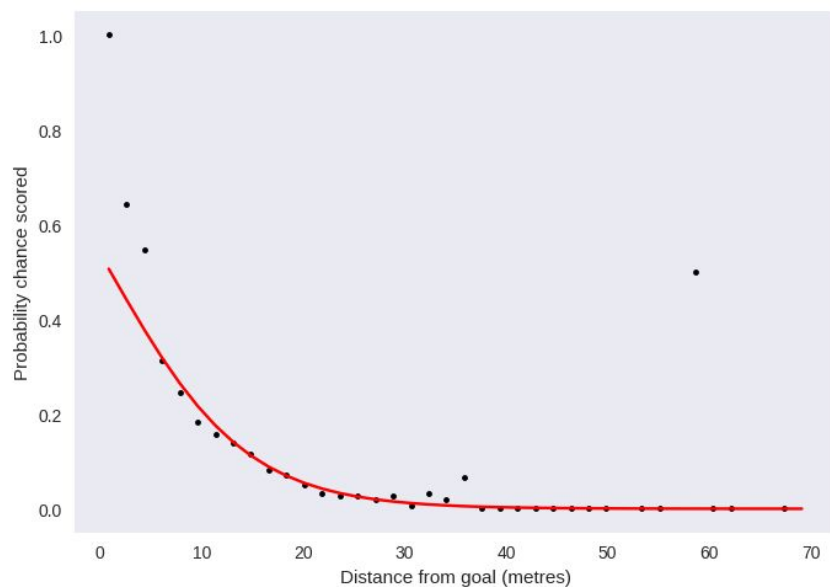


Modeling with distance

With logistic regression

Trained parameters

$b_0=0.1566$, $b_1=-0.1496$



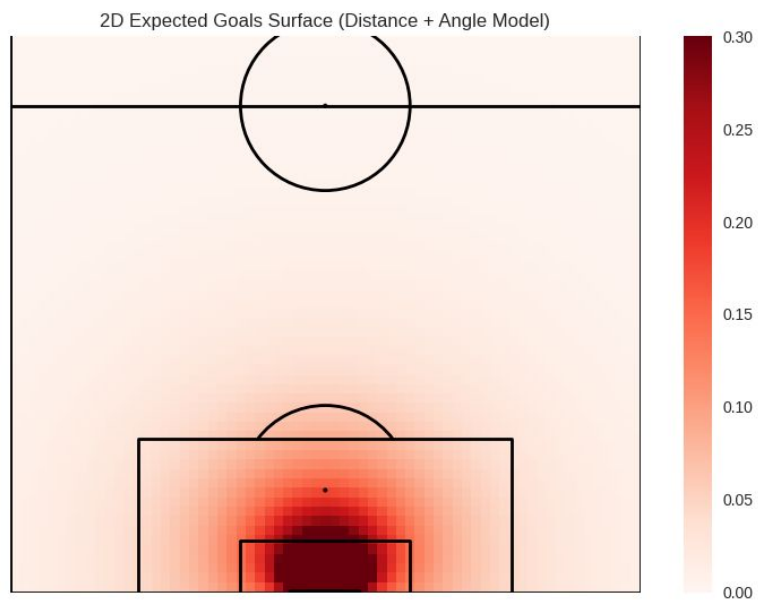
Modeling with angle & distance

With logistic regression

Trained parameters

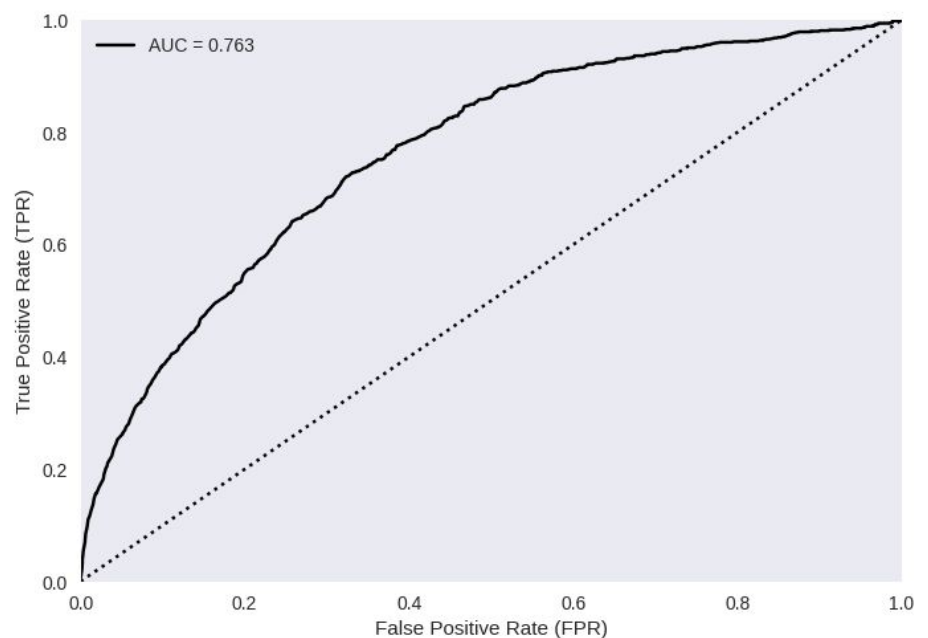
$b_0 = -1.1682$, $b_1 = -0.0994$,

$b_2 = 1.1467$

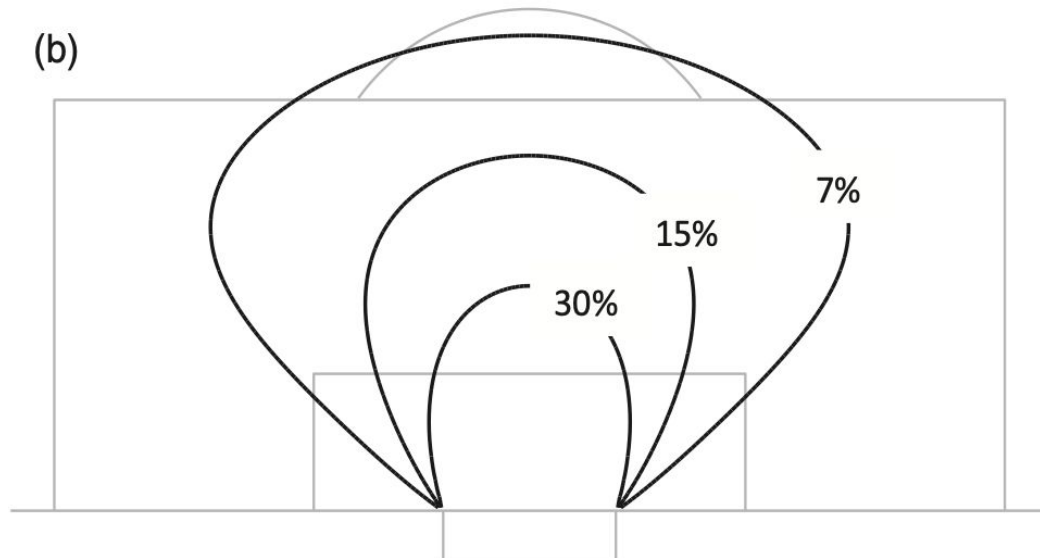


Modeling with angle & distance

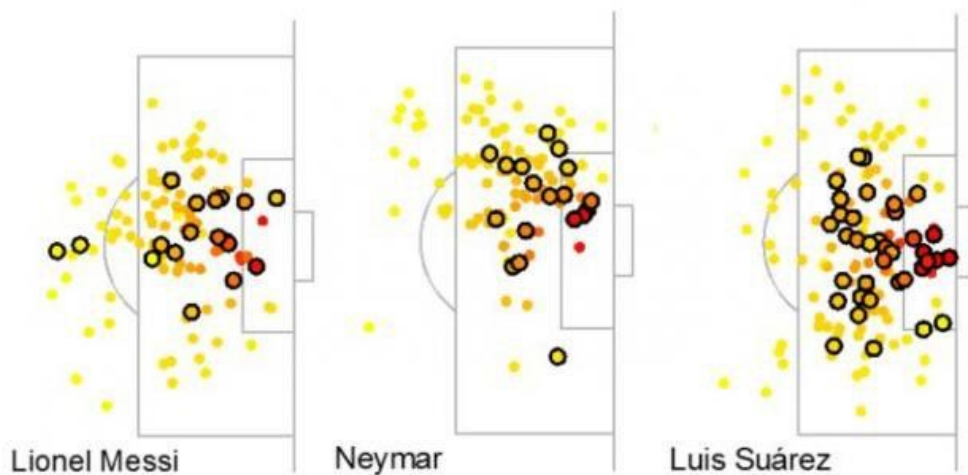
What do AUC tell us about?



Modeling with angle & distance



Shot & goal maps of different players



What can you say about their xG of different kind of shots like long distance, inside the box etc?

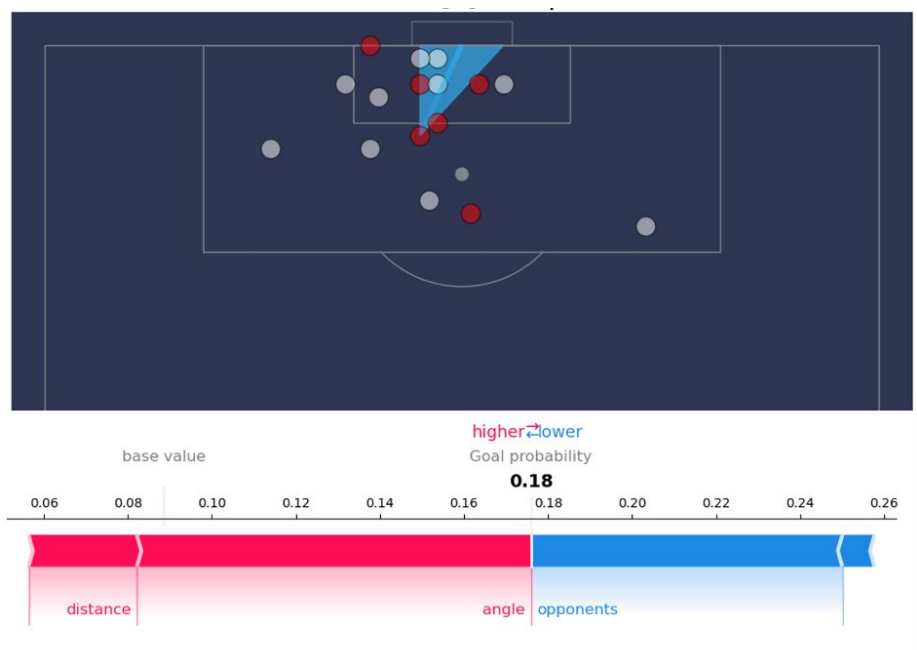
Shot & goal maps of different players



What can you say about their xG of different kind of shots like long distance, inside the box etc?

Other contributing factors

- What features have positive impact in xG here?
- What features have positive impact in xG here?



Other contributing factors

- What features have positive impact in xG here?
- What features have negative impact in xG here?

